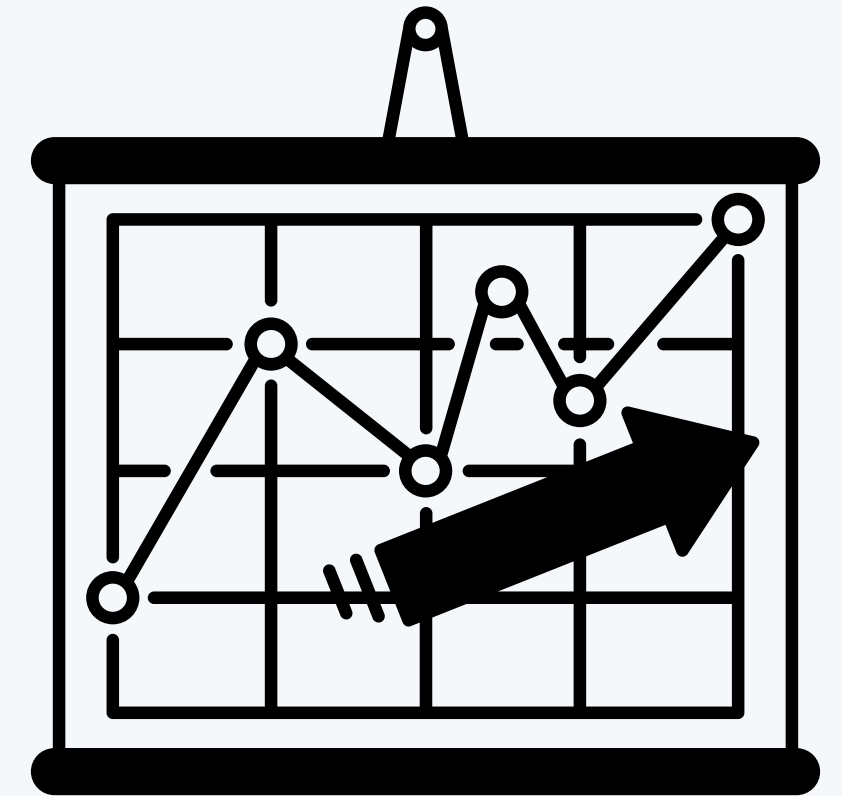


MemoryWatch

Quantum-Inspired Anomaly Detection for
Memory-Based Intrusion Detection Systems



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The Problem

Your passwords are in RAM right now and most tools cannot see them.

97.92%

Heap spray success rate on
Linux - Lee et al., 2023

Fileless

/proc injection leaves zero disk trace -
Nasereddin and Al-Qassas, 2024

Blind Spot

Signature tools are built for files. Memory attacks leave
none - Dehfouli and Habibi Lashkari, 2025

What MemoryWatch Does

One system.
Four components working together.

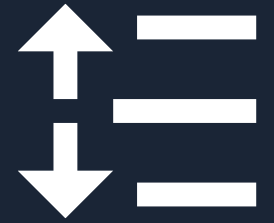
KERNEL MONITORING

Watches /proc access patterns and syscall frequency in real time



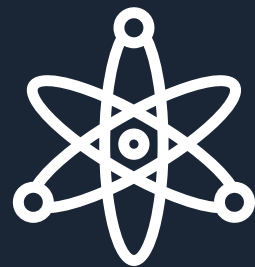
ISOLATION FOREST

Unsupervised detection. No labelled attack data needed. Dynamic thresholds per process.



QUANTUM KERNEL

ZZFeatureMap and one-class SVM. Better multiclass separation in feature space.



HONEST EVALUATION

FPR, Recall, F1, AUROC. Not just accuracy.



Threat Model - Three Attack Types

01 - Heap Spraying Lee et al., 2023

Attacker floods the heap with crafted objects. Forces a predictable layout. A buffer over-read overwrites a function pointer and hijacks execution. PSPRAY breaks SLUB randomisation using timing. Success rate: 97.92% across 10 real CVEs.

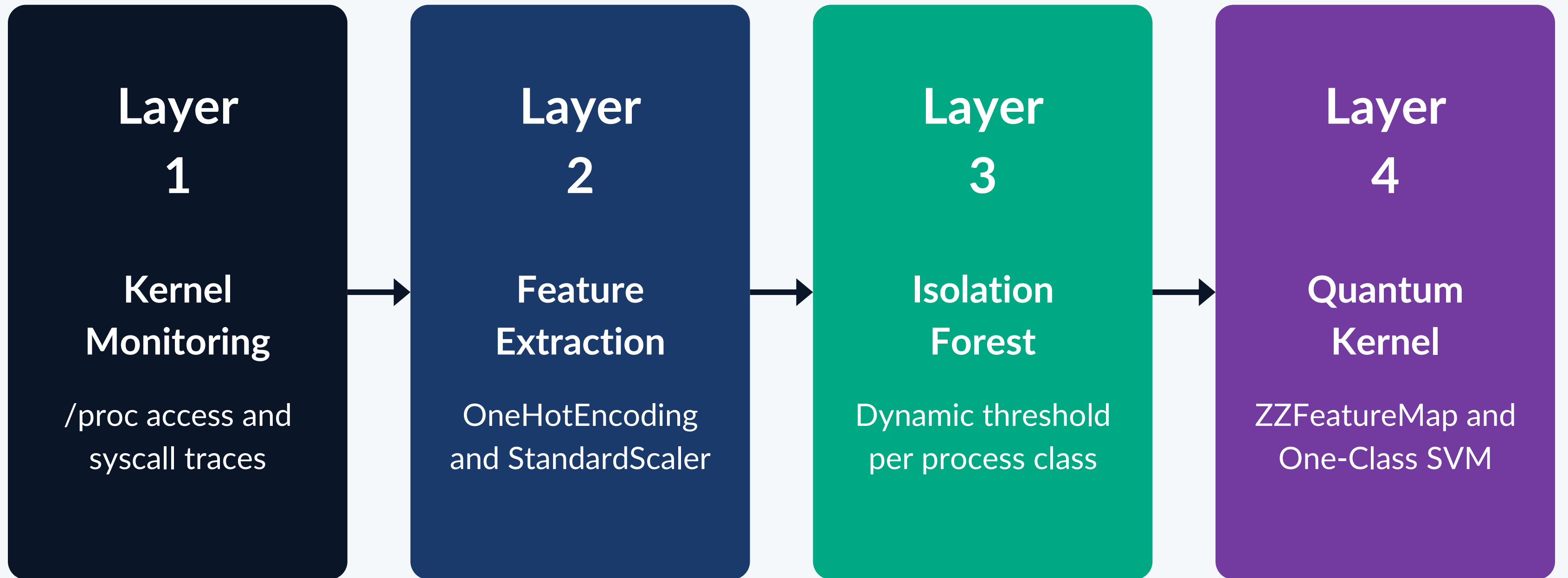
02 - /proc Injection Nasereddin & Al-Qassas, 2024

Every Linux process has a folder at /proc/[pid]. The file /proc/[pid]/mem exposes live memory. Attacker opens it like a normal file. Reads or writes directly. No new process. No binary. No disk trace.

03 - Memory-Resident Malware Dehfouli & Habibi Lashkari, 2025

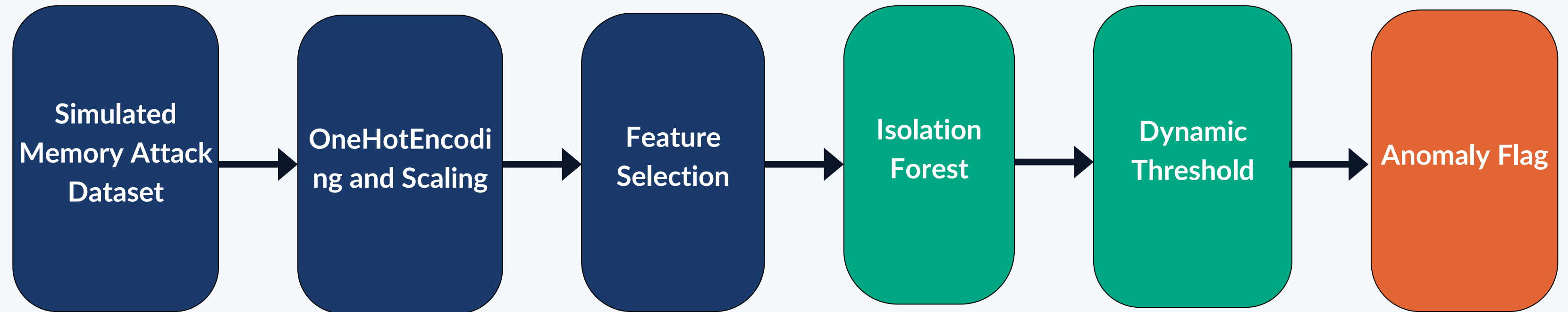
Malware lives entirely inside volatile memory. Static analysis, signature matching, heuristic scanning all fail. Live memory analysis is the only option.

System Architecture



Evaluation Output: FPR | Recall | F1 | AUROC

Classical ML Pipeline

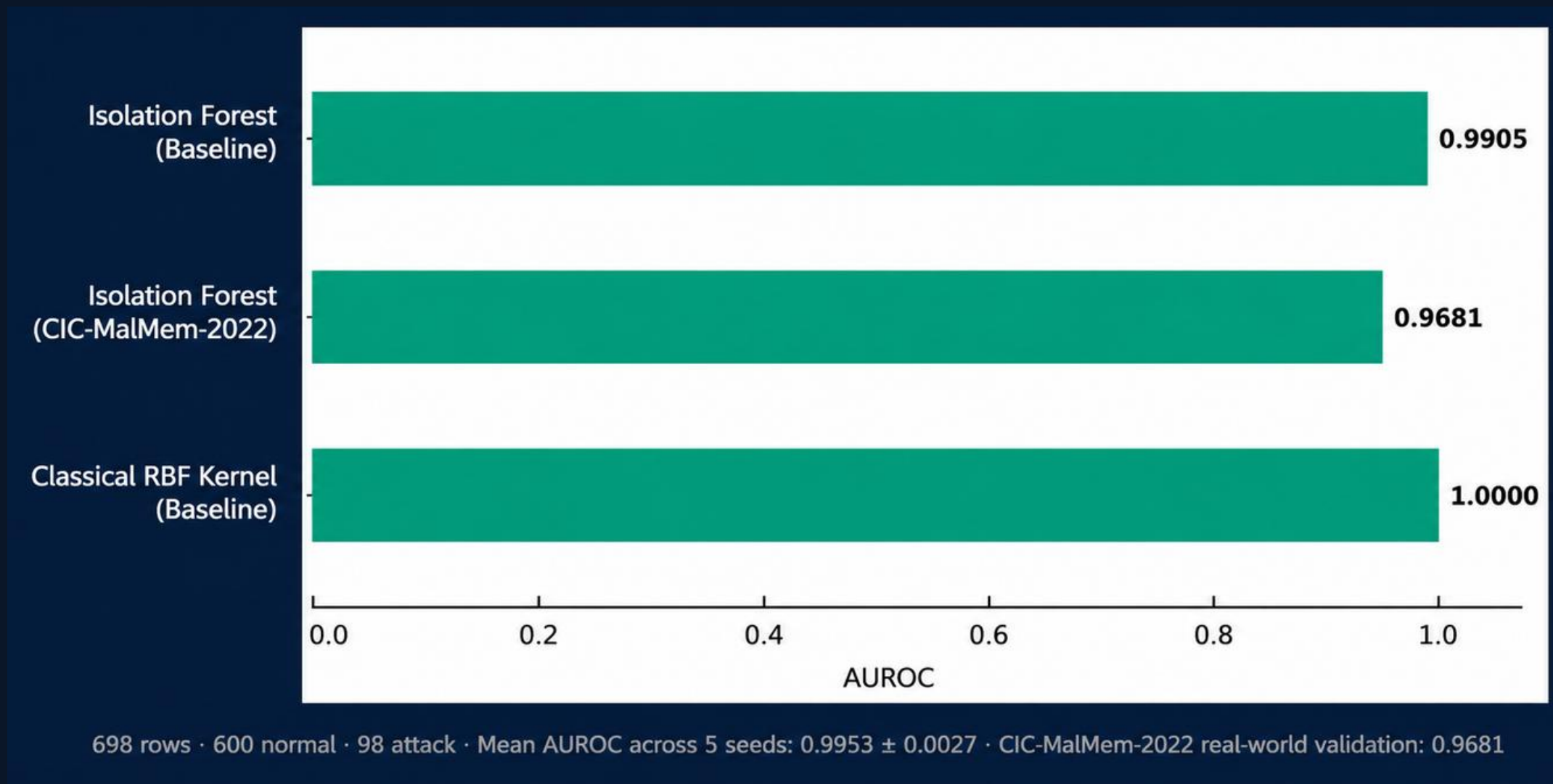


AUROC: 0.9953 (mean
across 5 seeds)
Std: ± 0.0027
CIC-MalMem-2022
validation: 0.9681

Per-process
threshold, not a fixed
global value -
Shamim et al., 2023

Two datasets used: Simulated
baseline (698 rows, 600 normal, 98
attack) + CIC-MalMem-2022 real-
world validation (58,596 rows,
29,298 benign, 29,298 malware)

Classical Results - AUROC Comparison

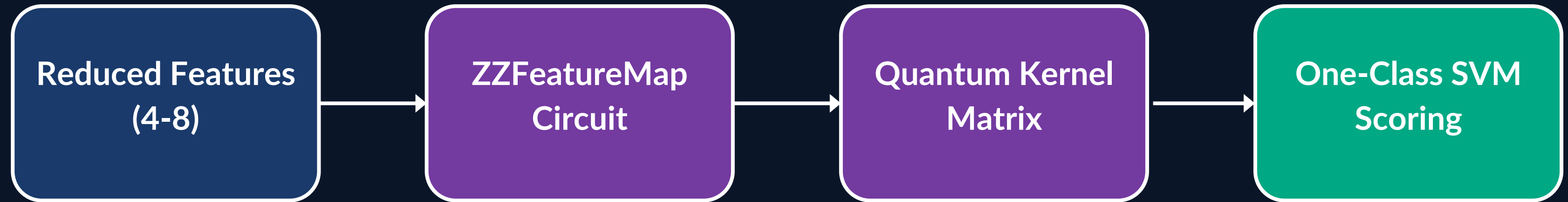


Important Context

These are benchmark numbers on the synthetic baseline dataset and CIC-MalMem-2022 only. Not real-world production deployment claims. Results reflect controlled experimental conditions. Not real-world deployment claims. Dataset has class imbalance and generalisability gaps. No single IDS outperforms across all datasets. Goldschmidt and Chuda, 2025 / Hesford et al., 2024

Quantum Layer

*Runs on Qiskit Statevector
Simulator*



Why quantum?

Features get mapped into a higher-dimensional Hilbert space. Attack classes that are not separable classically can become separable there.

**Where does it
actually help?**

Multiclass classification. Separating heap spray vs /proc injection vs credential scraping as distinct types.

Training limit

Training capped at 300 samples for kernel computation due to circuit complexity. Best quantum config found: 8 qubits, 2 reps, AUROC 0.9907.

Quantum Limitations

Why quantum is research, not deployment yet

GATE NOISE

Two-qubit gates have 0.1 to 1% error per operation. Errors build up with circuit depth - Preskill, 2018

SCALE CONSTRAINTS

AUROC was 0.8369 at 4 qubits & 0.9907 at 8 qubits. More qubits help but current hardware cannot reach this scale. Training capped at 300 samples.

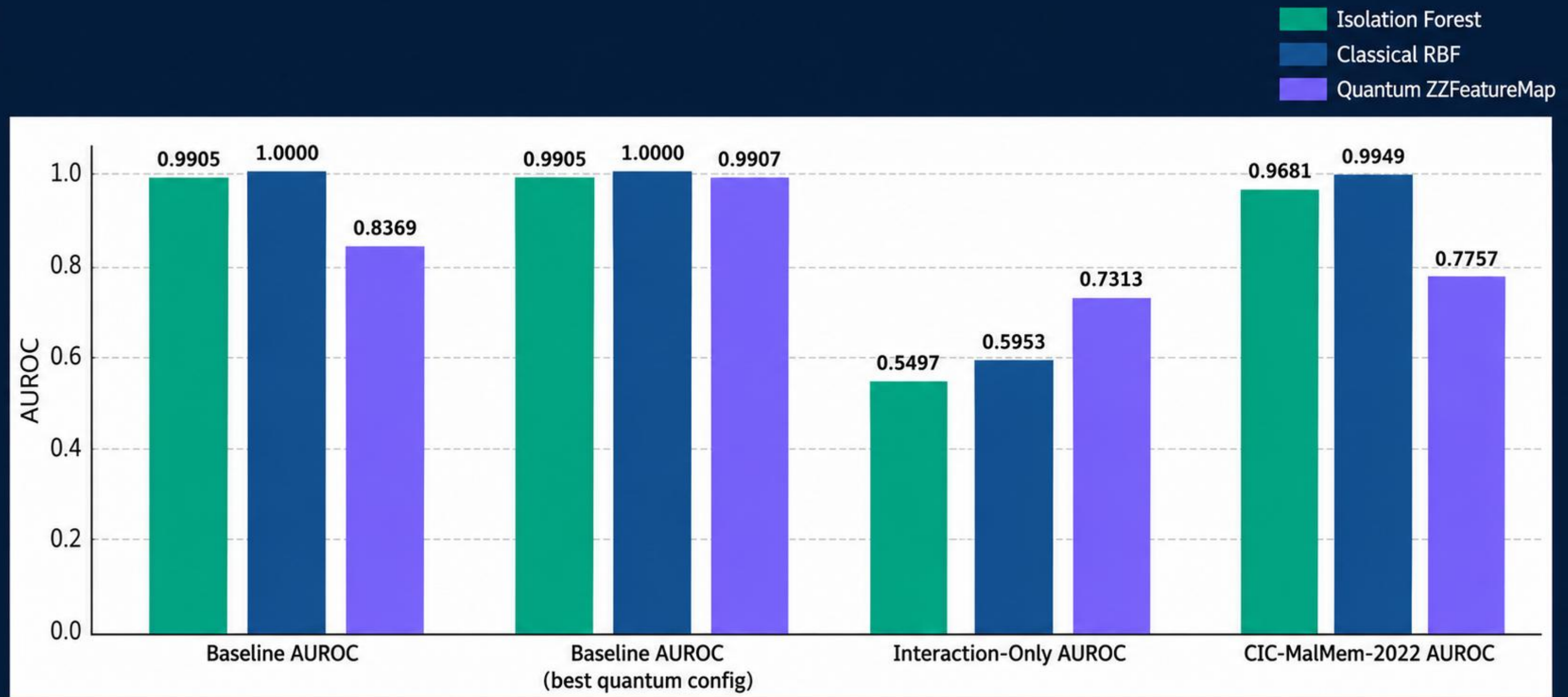
SIMULATION IS NOT HARDWARE

Qiskit runs on a classical CPU. No gate noise. No decoherence. Real IBM hardware means hours of queuing.

ENTERPRISE INFEASIBLE TODAY

Cryogenic cooling. Cloud access. Hours of queuing. Real-time IDS needs milliseconds.

Classical vs Quantum



Quantum 4q reps=1 on baseline: 0.8369. Best config 8q reps=2: 0.9907. Interaction-only quantum won 4/5 seeds. Mean advantage: +0.0337.

Evaluation and Dataset

Why not just accuracy?

86% false alarm rate makes analysts 47% less precise and 40% slower - Layman and Roden, 2023

F1 ranged from 0.02 to 0.95 for the same IDS across different datasets - Hesford et al., 2024

Primary metrics: FPR, Recall, F1, AUROC

Accuracy is supplementary only

Datasets Used

Simulated baseline: 698 rows, 600 normal, 98 attack

3 attack types: heap spray, mem injection, proc scraper

CIC-MalMem-2022: 58,596 rows, 29,298 benign, 29,298 malware

3 malware families: Ransomware, Spyware, Trojan

Results reported as benchmark only

Not real-world deployment claims

Output Runs

```
(root@ kali)-[~/code]
# python3 memorywatch_final.py
#####
MEMORYWATCH – FINAL CONSOLIDATED QUANTUM VS CLASSICAL PIPELINE
#####

[step 1/8] Generating baseline dataset...
698 rows (600 normal, 98 attack)

[step 2/8] Generating interaction-only dataset...

Design verification – single-feature AUROC (should be ~0.5):
syscall_entropy      0.500  ✓ ok
fd_count             0.535  ✓ ok
maps_exec_region_count 0.523  ✓ ok

642 rows (600 normal, 42 attack)

[step 3/8] Dataset QA...

Dataset QA: baseline

600 normal | 98 attack | ratio=14.0%
Strongest single features: syscall_entropy, exec_region_growth_rate, maps_anonymous_exec (AUROC 1.000, 0.857, 0.849)
Weakest single features:  maps_total_regions, maps_exec_region_count, io_write_bytes_delta (AUROC 0.549, 0.623, 0.639)
△ 6 feature(s) fully inside normal range (no standalone signal): ['vmrss_kb', 'vmrss_delta', 'fd_count', 'maps_total_regions', 'maps_exec_region_count', 'io_write_bytes_delta']

Dataset QA: interaction-only

600 normal | 42 attack | ratio=6.5%
△ Zero-variance features: ['fd_proc_mem_open', 'exec_region_growth_rate']
Strongest single features: io_read_bytes_delta, fd_count, vmrss_kb (AUROC 0.551, 0.535, 0.529)
Weakest single features:  syscall_entropy, fd_proc_mem_open, exec_region_growth_rate (AUROC 0.500, 0.500, 0.500)
△ 11 feature(s) fully inside normal range (no standalone signal): ['vmrss_kb', 'vmrss_delta', 'fd_count', 'fd_proc_mem_open', 'maps_total_regions', 'maps_exec_region_count', 'maps_anonymous_exec', 'exec_region_growth_rate', 'syscall_entropy', 'io_read_bytes_delta', 'io_write_bytes_delta']

[step 4/8] Three-way comparison (default config: 4 qubits, reps=1)...
```


Three-way comparison: baseline dataset

Isolation Forest (weighted=True, f1-optimal threshold)

AUROC=0.9905

heap_spray	42/42	100.0%
mem_injection	28/28	100.0%
proc_scraper	27/28	96.4%

PCA(4) explained variance: 89.1%

Classical RBF Kernel AUROC=1.0000

heap_spray	42/42	100.0%
mem_injection	28/28	100.0%
proc_scraper	28/28	100.0%

Quantum ZZFeatureMap Kernel (qubits=4, reps=1) AUROC=0.8369

heap_spray	29/42	69.0%
mem_injection	16/28	57.1%
proc_scraper	15/28	53.6%

Summary: IF=0.9905 RBF=1.0000 Quantum=0.8369 (Quantum - RBF = -0.1631)

Three-way comparison: interaction-only dataset

Isolation Forest (weighted=True, f1-optimal threshold)

AUROC=0.5497

interaction_only	31/42	73.8%
------------------	-------	-------

PCA(4) explained variance: 89.1%

Classical RBF Kernel AUROC=0.5953

interaction_only	12/42	28.6%
------------------	-------	-------

Quantum ZZFeatureMap Kernel (qubits=4, reps=1) AUROC=0.5864

interaction_only	19/42	45.2%
------------------	-------	-------

Summary: IF=0.5497 RBF=0.5953 Quantum=0.5864 (Quantum - RBF = -0.0089)

[step 5/8] Multi-seed validation (baseline dataset)...

```
=====
Multi-seed validation – baseline dataset, seeds=[42, 7, 99, 123, 2026]
=====
seed=42    IF=0.9905  RBF=1.0000  Quantum=0.8369
seed=7     IF=0.9962  RBF=1.0000  Quantum=0.7832
seed=99    IF=0.9975  RBF=1.0000  Quantum=0.7325
seed=123   IF=0.9979  RBF=1.0000  Quantum=0.7982
seed=2026  IF=0.9942  RBF=1.0000  Quantum=0.8452

Method          mean ± std AUROC
isolation_forest 0.9953 ± 0.0027
rbf_kernel       1.0000 ± 0.0000
quantum_kernel   0.7992 ± 0.0406

[step 6/8] Qubit x reps sweep...
```

```
=====
Qubit x reps sweep – baseline dataset
=====
qubits  reps  expl.var    RBF    Quantum    Δ
    4     1         1.0000    0.8369  -0.1631
    4     2         1.0000    0.8019  -0.1981
    6     1         1.0000    0.9395  -0.0605
    6     2         1.0000    0.9265  -0.0735
    8     1         1.0000    0.9783  -0.0217
    8     2         1.0000    0.9907  -0.0093

Best quantum config: qubits=8, reps=2 -> AUROC=0.9907 (RBF at same config: 1.0000)
```


=====

Qubit x reps sweep – interaction-only dataset

=====

qubits	reps	expl.var	RBF	Quantum	Δ	
4	1		0.5953	0.5864	-0.0089	
4	2		0.5953	0.5954	+0.0001	<-- quantum wins
6	1		0.6395	0.6406	+0.0011	<-- quantum wins
6	2		0.6395	0.6969	+0.0574	<-- quantum wins
8	1		0.7379	0.6730	-0.0649	
8	2		0.7379	0.7313	-0.0065	

Best quantum config: qubits=8, reps=2 -> AUROC=0.7313 (RBF at same config: 0.7379)

[step 7/8] Seed-locked validation of best quantum config...

=====

Seed-locked validation: interaction dataset, qubits=8, reps=2, seeds=[42, 7, 99, 123, 2026]

=====

(Checks whether the best config found by the sweep is a real effect or noise from testing multiple configs and picking one.)

seed=42	RBF=0.7379	Quantum=0.7313	$\Delta=-0.0065$	RBF wins/ties
seed=7	RBF=0.6884	Quantum=0.7611	$\Delta=+0.0727$	QUANTUM WINS
seed=99	RBF=0.7671	Quantum=0.7849	$\Delta=+0.0179$	QUANTUM WINS
seed=123	RBF=0.6842	Quantum=0.7399	$\Delta=+0.0557$	QUANTUM WINS
seed=2026	RBF=0.7123	Quantum=0.7413	$\Delta=+0.0289$	QUANTUM WINS

Quantum won on 4/5 seeds.

Mean Δ (Quantum - RBF) = $+0.0337 \pm 0.0279$

VERDICT: quantum advantage appears REAL and reasonably consistent at this configuration on the interaction-only dataset.

[step 8/8] Real-world validation (CIC-MalMem-2022, auto-discovered)...

Loading /root/code/Obfuscated-MalMem2022.csv ...

58596 rows, 57 columns

Detected label column(s): class='Class', category='Category'

Using 55 numeric feature columns

Dropping 3 zero-variance columns: ['pslist.nprocs64bit', 'handles.nport', 'svcsan.interactive_process_services']

29298 benign, 29298 malware in full dataset

Subsampled to 3000 rows (1500 benign, 1500 malware) for kernel computation

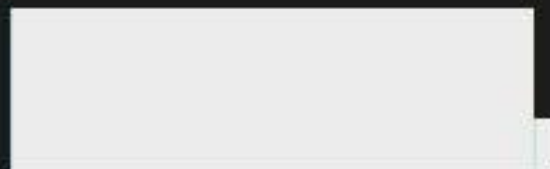
Malware family breakdown in subsample: {'Ransomware': 519, 'Trojan': 478, 'Spyware': 503}


```
=====
Three-way comparison: Obfuscated-MalMem2022.csv
=====
```

Isolation Forest (unweighted – no domain priors for this feature set)

AUROC=0.9681

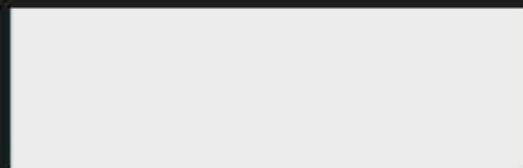
Ransomware	518/519	99.8%
Spyware	496/503	98.6%
Trojan	478/478	100.0%



PCA(6) explained variance: 96.5% (kernel train set: 300 normal rows)

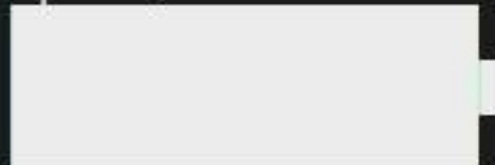
Classical RBF Kernel AUROC=0.9949

Ransomware	518/519	99.8%
Spyware	499/503	99.2%
Trojan	477/478	99.8%



Quantum ZZFeatureMap Kernel (qubits=6, reps=2) AUROC=0.7757

Ransomware	467/519	90.0%
Spyware	469/503	93.2%
Trojan	428/478	89.5%



Summary: IF=0.9681 RBF=0.9949 Quantum=0.7757 (Quantum - RBF = -0.2192)

```
#####
```

DONE in 32.2s

```
#####
```

```
(root@kali)-[~/code]
#
```


USP

Three gaps, One system and MemoryWatch addresses all three.

01

No mainstream tool detects fileless memory attacks in real time without signatures.

02

No classical IDS pipeline built specifically for memory-based attack features with dynamic thresholds

03

No study has run this comparison experimentally. MemoryWatch did across 5 seeds, two datasets and a qubit sweep from 4 to 8

Conclusion

What was proved?

Memory-based attacks are real, actively exploited and invisible to signature tools.

Isolation Forest with dynamic thresholds is validated, practical and gives low FPR

Quantum kernels showed a real and consistent advantage on 4 out of 5 seeds on the interaction-only dataset at 8 qubits, 2 reps. Mean AUROC gain: +0.0337.

Quantum deployment is not feasible today. The research case is proven.

Thank You



for your time and attention